

A Complete Cycle Atlas for Projective Möbius Dynamics over Finite Fields

Route-A source-local certificate HCS-C260

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Abstract

For every prime power $q = p^r$, we classify the permutation induced by each $g \in \mathrm{PGL}_2(\mathbb{F}_q)$ on $\mathbb{P}^1(\mathbb{F}_q)$. Four cases give the complete cycle multiset, all iterated fixed counts, primitive cycles, the finite Artin–Mazur zeta, and the finite Koopman spectrum. Exhaustive direct permutation checks cover 18 prime and extension fields. We also close the characteristic-two classifier, the exact per-order type census, and explicit projective reversors in every characteristic. All statements are source-local and use no target spectral data.

1 The four cycle types

Represent g by $A \in \mathrm{GL}_2(\mathbb{F}_q)$, modulo nonzero scalar. In the semisimple cases let d be the order of the eigenvalue ratio.

Theorem 1 (complete projective cycle atlas). *Exactly one of the following rows applies.*

<i>type</i>	<i>condition</i>	<i>cycles on $\mathbb{P}^1(\mathbb{F}_q)$</i>	<i>$F_n = \# \mathrm{Fix}(g^n)$</i>
<i>identity</i>	$g = 1$	1^{q+1}	$q + 1$
<i>unipotent</i>	<i>nontrivial, order p</i>	$1^1 p^{q/p}$	$1 + q \mathbf{1}_{p n}$
<i>split</i>	$d > 1, d \mid q - 1$	$1^2 d^{(q-1)/d}$	$2 + (q - 1) \mathbf{1}_{d n}$
<i>nonsplit</i>	$d > 1, d \mid q + 1$	$d^{(q+1)/d}$	$(q + 1) \mathbf{1}_{d n}$

For every $m \geq 1$, the numbers of points and cycles of least period m are

$$P_m = \sum_{e|m} \mu(m/e) F_e, \quad C_m = P_m/m. \quad (1)$$

Proof. The scalar case is immediate. A nonscalar repeated-root matrix has one rational eigenline and is projectively conjugate to $x \mapsto x + 1$. It fixes infinity and partitions the affine line into q/p cycles of length p . If the characteristic polynomial splits with distinct roots, conjugate the two rational eigenlines to 0 and infinity. The map is $x \mapsto \rho x$; ρ has order $d \mid q - 1$ and acts freely on $\mathbb{F}_q^*$.

In the irreducible case the two eigenlines lie over $\mathbb{F}_{q^2}$ and Frobenius interchanges them. Thus $\rho^q = \rho^{-1}$ and $d \mid q + 1$. A proper nonidentity power cannot acquire a rational eigenline: it retains the same two nonrational eigenlines. Hence the cyclic action is semiregular on all $q + 1$ rational projective points. The fixed formulas sum cycle lengths dividing n , and Möbius inversion gives (1). $\square$

2 Characteristic two and the type census

When $p = 2$, discriminant-square language is not a valid classifier. If $\mathrm{tr} A = 0$, the characteristic polynomial has a repeated root; apart from the scalar identity this is the unipotent face. If $\mathrm{tr} A \neq 0$, scale the polynomial to

$$y^2 + y + u, \quad u = \frac{\det A}{(\mathrm{tr} A)^2}. \quad (2)$$

The Artin–Schreier equation $y^2 + y = u$ is soluble in $\mathbb{F}_q$ precisely when $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(u) = 0$. Absolute trace zero therefore means split, while absolute trace one means nonsplit.

The order-refined census is also exact. There is one identity and $q^2 - 1$ nonidentity unipotents. For every $d > 1$,

$$\begin{aligned} d \mid q - 1 : \quad & \#\{\text{split elements of order } d\} = q(q + 1)\varphi(d)/2, \\ d \mid q + 1 : \quad & \#\{\text{nonsplit elements of order } d\} = q(q - 1)\varphi(d)/2. \end{aligned} \quad (3)$$

For $d > 2$, pair the $\varphi(d)$ ratios as $\{\rho, \rho^{-1}\}$ and use torus centralizers of sizes $q - 1$ and $q + 1$. At $d = 2$ the ratio is self-inverse and its torus normalizer doubles the centralizer, producing the same formulas. This boundary is important for odd q : split involutions have two rational fixed points, whereas nonsplit involutions have none. In characteristic two, $q - 1$ and $q + 1$ are odd, so nonidentity projective involutions are unipotent. Summing the census with $\sum_{d|N} \varphi(d) = N$ gives $q(q^2 - 1)$.

3 Uniform exact reversibility

Every nonscalar 2×2 matrix is cyclic. In a cyclic basis put

$$C = \begin{pmatrix} 0 & -\delta \\ 1 & \tau \end{pmatrix}, \quad H = \begin{pmatrix} 0 & \delta \\ 1 & 0 \end{pmatrix}, \quad \tau = \text{tr } A, \quad \delta = \det A. \quad (4)$$

Direct multiplication, with no characteristic restriction, yields

$$H^2 = \delta I, \quad HCH^{-1} = \delta C^{-1}. \quad (5)$$

Thus $[H]$ is a projective involution and $[H]g[H]^{-1} = g^{-1}$. The identity is trivially reversible. This closes the reversor problem for all four rows, including characteristic two; it does not assert that the Koopman operator itself is self-adjoint (that occurs only when the projective order is at most two).

4 Zeta and finite Koopman closure

For a finite permutation with c_L cycles of length L ,

$$\zeta_g(t) = \exp\left(\sum_{n \geq 1} \frac{F_n}{n} t^n\right) = \prod_L (1 - t^L)^{-c_L}. \quad (6)$$

Consequently the four rows give, respectively,

$$(1 - t)^{-(q+1)}, \quad (1 - t)^{-1}(1 - t^p)^{-q/p}, \quad (1 - t)^{-2}(1 - t^d)^{-(q-1)/d}, \quad (1 - t^d)^{-(q+1)/d}. \quad (7)$$

On counting-measure $\ell^2(\mathbb{P}^1(\mathbb{F}_q))$, the Koopman operator is a permutation unitary. Every L -cycle contributes each L th root of unity once, and

$$\det(I - tU_g) = \prod_L (1 - t^L)^{c_L} = \zeta_g(t)^{-1}. \quad (8)$$

5 Exact regression and boundary

Two implementations cover $q = 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32$. The independent checker shares no finite-field or classifier code with the producer: it rebuilds every field and projective permutation, derives the type only from fixed-point geometry and cycle lcm, and matches a canonical per-matrix record hash. In total it checks 155,346 transformations, 4,367,094 state images, and closes 6,159,318 assertions. The extension-field sentinels have degrees two through five and exercise the absolute-trace rule. Symbolic algebra closes 193 identities; byte replay is exact; semantic mutation rejects 40/40 repaired-hash changes, including swapped $q - 1/q + 1$ channels and corrupted involution faces. Finite enumeration is a regression oracle, not the proof for arbitrary q .

Finite-field arithmetic provides only `A0_WEAK_ARITHMETIC_RELATION`; Theorem 1 earns `A1_PASS_ANALYTIC`, while (8) gives `A4_NATURAL_QUANTIZATION`. There is no rational-prime orbit dictionary, target divisor, or target global analytic structure. The strict tuple is

$$(\text{A0_WEAK_ARITHMETIC_RELATION}, \text{A1_PASS_ANALYTIC}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_NATURAL_QUANTIZATION}).$$

The verdict is `ROUTE_A_EXPLORATORY`; Route B is disabled. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, (6) is only a finite source zeta, and (8) is not a Hilbert–Polya identification. We claim neither literature priority for the classical context in Refs. [1–3] nor that finite tests prove the arbitrary-prime-power theorem.

References

- [1] E. Sakzad, M.-R. Sadeghi, and D. Panario, *Adv. Math. Commun.* **6** (2012), 347–361, [doi:10.3934/amc.2012.6.347](https://doi.org/10.3934/amc.2012.6.347).
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